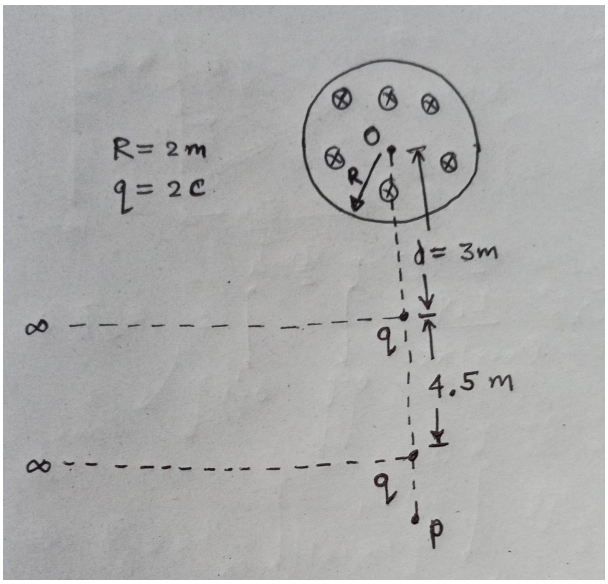
PH1201: Electricity and Magnetism

Tutorial Session-10

April 23, 2024

Problem 1: More on Quasi-static condition

An infinitely long straight wire carries a slowly varying current $I(t)$. Determine the induced electric field, as a function of the distances from the wire.

Problem 2: Graded

The magnetic field in the cylindrical region of radius 2 m is changing with time as $\frac{dB}{dt} = 2Ts^{-1}$. A positive point charge of magnitude, 2 C is carried by an external force from the line OP all the way up to infinity along the two paths characterised by the their distance from O as shown in the figure. The charge is carried in such way that no kinetic energy is gained by the charge at any moment in its journey. Find the work done by the external agent (along with its sign) along the two paths.